



KTU NOTES

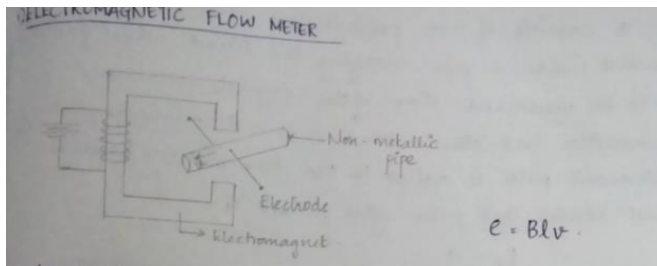
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Part A

1. Explain the working of the electromagnetic flow meter. [3 marks] (December 2021)



→ Pipe carrying the liquid whose flow rate is to be measured is placed under an electromagnet. The velocity of flow of liquid be ' v '. Since, liquid is of conducting nature, it acts as a moving conductor placed in $\vec{B}$. Hence, an emf will be induced across the 2 electrodes connected to the pipe. The pipe should be of Non-metallic type.

2 Explain any three types of oscilloscope probes. [3 marks] (December 2021)

Types of CRO Probes

CRO probes can be classified into the following **two types**.

- ▣ Passive Probes
- ▣ Active Probes

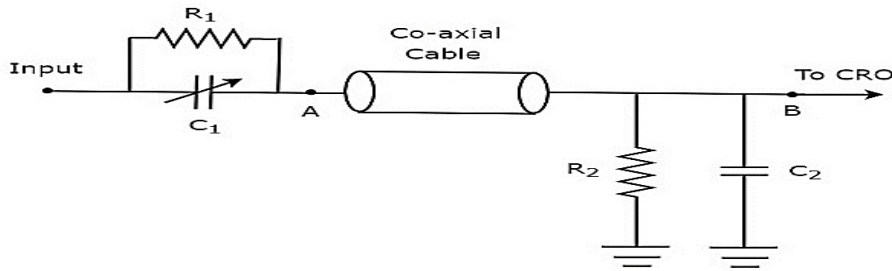
Now, let us discuss about these two types of probes one by one.

Passive Probes

If the probe head consists of passive elements, then it is called **passive probe**. The circuit diagram of passive probe is shown in below figure.

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As shown in the figure, the probe head consists of a parallel combination of resistor, R_1 and a variable capacitor, C_1 . Similarly, the termination circuit consists of a parallel combination of resistor, R_2 and capacitor, C_2 .

Active Probes

Active probe differs from the passive probes because the probe tip contains a small active amplifier built into the probe body near the probe tip so that the probe input capacitance can be kept very small, usually less than 2pF.

This low capacitance results in high input impedance on high frequencies.

It has the best overall combination of resistive and capacitive loading.

With such low loading, active probes can be used on high-impedance circuits that would be seriously loaded by passive probes.

Tri- Mode Probes

- Tri-Mode Probes are a relatively new probe type that provide the advantage of being able to use one setup, and get three types of measurements without adjusting probe tip connections.
- The Tri-Mode probe can make differential, single-ended and common mode measurements from the same probe setup.

3. What is lissajous pattern and how does it vary with phase shift.

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[3 marks] (December 2020)

How frequency and phase is measured using Lissajous pattern. [4 marks] (December 2020)

Measurement of Phase and Frequency (Lissajous Patterns).

the characteristics of patterns that appear on the screen of a CRT when sinusoidal voltages are simultaneously applied to horizontal and vertical plates. are called 'Lissajous Patterns'.

When two sinusoidal voltages of equal frequency which are in phase with each other are applied to the horizontal and vertical deflection plates, the pattern appearing on the screen is a straight line as is clear from Fig.

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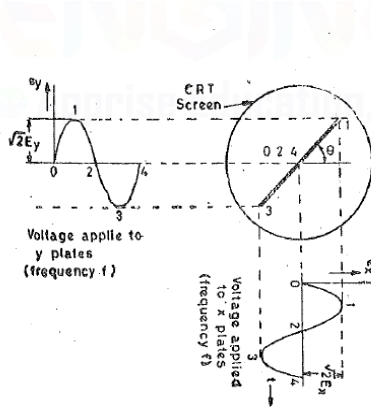


Fig. Lissajous pattern with equal frequency voltages and zero phase shift.

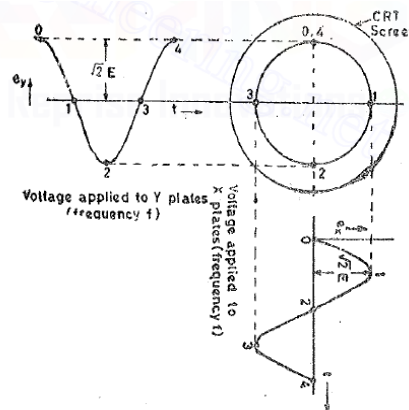


Fig. Lissajous pattern with equal voltages of equal frequency and a phase shift of 90° .

Thus when two equal voltages of equal frequency but with 90° phase displacement are applied to a CRO, the trace on the screen is a circle. This is shown in Fig.

When two equal voltages of equal frequency but with a phase shift ϕ (not equal to 0° or 90°) are applied to a CRO we obtain an ellipse as shown in Fig. An ellipse is also obtained when unequal voltages of same frequency are applied to the CRO.

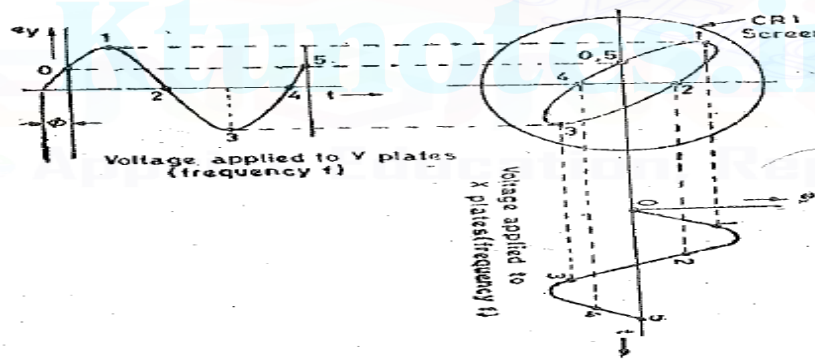
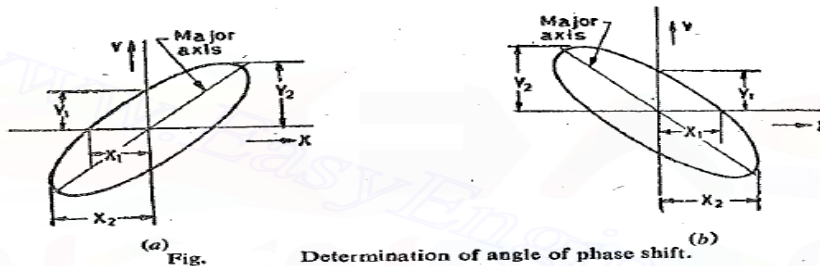


Fig. Lissajous pattern with two equal voltages of same frequency and phase shift of ϕ .



(a) Fig. Determination of angle of phase shift. (b)

Regardless of the two amplitudes of the applied voltages the ellipse provides a simple means of finding phase difference between two voltages. Referring to Fig. , the sine of the phase angle between the voltages is given by :

$$\sin \phi = \frac{Y_1}{Y_2} = \frac{X_1}{X_2}$$

For convenience, the gains of the vertical and horizontal amplifiers are adjusted so that the ellipse fits exactly into a square marked by the lines on the graticule.

If the major axis of the ellipse lies in the first and third quadrants (i.e., its slope is positive) as in Fig. (a), the phase angle is either between 0° to 90° or between 270° to 360° . When the major axis of ellipse lies in second and fourth quadrants i.e., when its slope is negative as in Fig. (b), the phase angle is either between 90° and 180° or between 180° and 270° .

Frequency Measurements. Lissajous patterns may be used for accurate measurement of frequency. The signal, whose frequency is to be measured, is applied to the Y plates. An accurately calibrated standard variable frequency source is used to supply voltage to the X plates, with the internal sweep generator switched off. The standard frequency is adjusted until the pattern appears as a circle or an ellipse, indicating that both signals are of the same frequency. Where it is not possible to adjust the standard signal frequency to the exact frequency of the unknown signal, the standard is adjusted to a multiple or a submultiple of the frequency of the unknown source so that the pattern appears stationary.

The two waves start at the same instant. Lissajous pattern may be constructed in the usual way and a 8 shaped pattern with two loops is obtained. If the two waves do not start at the same instant we get different patterns for the same frequency ratio. The Lissajous patterns for other frequency ratios can be similarly drawn. Some of these patterns are shown in Fig.

It can be shown that for all the above cases, the ratios of the two frequencies is :

$$\frac{f_v}{f_x} = \frac{\text{number of times tangent touches top or bottom}}{\text{number of times tangent touches either side}} = \frac{\text{number of horizontal tangencies}}{\text{number of vertical tangencies}}$$

where
and

f_v = frequency of signal applied to Y plates.
 f_x = frequency of signal applied to X plates.

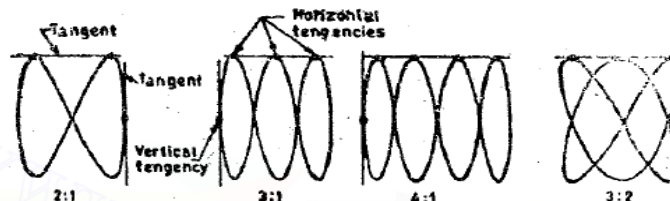


Fig. Lissajous patterns with different frequency ratios.

4. List the merits and demerits of LVDT. [3 marks] (December 2020)

- The most widely used inductive transducer to translate the linear motion into electrical signals is the Linear Variable Differential Transformer (LVDT).

Advantages

- Linearity is good up to 5 mm of displacement.
- Output is rather high. Therefore, immediate amplification is not necessary.

- Output voltage is stepless and hence the resolution is very good.
- Sensitivity is high (about 40 V/mm).
- It does not load the measurand mechanically.
- It consumes low power and low hysteresis loss also.

Disadvantages

- LVDT has large threshold.
 - It is affected by stray electromagnetic fields. Hence proper shielding of the device is necessary.
 - The ac inputs generate noise.
 - Its sensitivity is lower at higher temperature.
 - Being a first-order instrument, its dynamic response is not instantaneous.
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PART B

1. Explain the basic principle and working of LVDT. [6 marks] (December 2021)

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Briefly explain LVDT with neat figures. What are the merits and demerits of LVDT? Also list down any two applications. [10 marks] (December 2020)

- The most widely used inductive transducer to translate the linear motion into electrical signals is the Linear Variable Differential Transformer (LVDT).
- The basic construction of LVDT is shown in Figure .
- The transformer consists of a single primary winding 'P' and two secondary windings S1 and S2 wound on a cylindrical former.
- A sinusoidal voltage of amplitude 3 to 15 Volt and frequency 50 to 20000 Hz is used to excite the primary.
- The two secondaries have equal number of turns and are identically placed on either side of the primary winding.

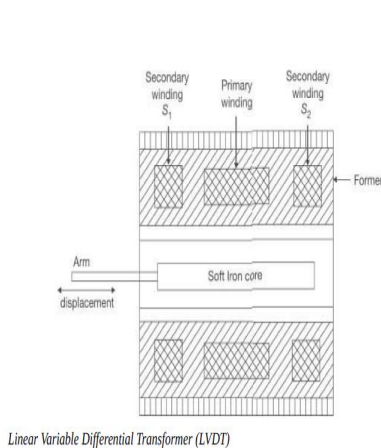


Figure (a)

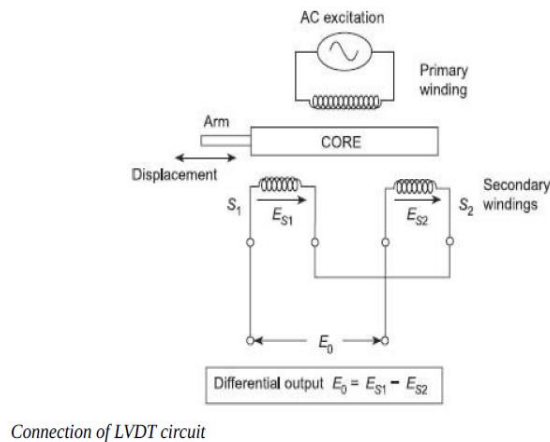


Figure (b)

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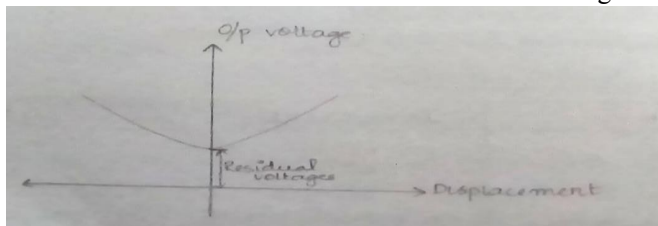
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- The primary winding is connected to an alternating current source.
- A movable soft-iron core is placed inside the former.
- The displacement to be measured is applied to the arm attached to the soft iron core.
- In practice the core is made of high permeability, nickel iron.
- This is slotted longitudinally to reduce eddy current losses.
- The assembly is placed in a stainless steel housing to provide electrostatic and electromagnetic shielding.
- The frequency of ac signal applied to primary winding may be between 50 Hz to 20 kHz.
- Since the primary winding is excited by an alternating current source, it produces an alternating magnetic field which in turn induces alternating voltages in the two secondary windings.
- The output voltage of secondary S_1 is ES_1 and that of secondary S_2 is ES_2 .
- In order to convert the outputs from S_1 and S_2 into a single voltage signal, the two secondary S_1 and S_2 are connected in series opposition as shown in Figure (b) .
- Differential output voltage $E_0 = ES_1 - ES_2$
- When the core is at its normal (NULL) position, the flux linking with both the secondary windings is equal and hence equal emfs are induced in them.
- Thus, at null position: $ES_1 = ES_2$. Thus, the output voltage E_0 is zero at null position.

Now if the core is moved to the left of the null position, more flux links with S_1 and less with winding S_2 . Accordingly, output voltages ES_1 is greater than ES_2 .

The magnitude of output voltage is thus, $E_0 = ES_1 - ES_2$ and say it is in phase with primary voltage.

- Similarly, when the core is moved to the right of the null position ES_2 will be more than ES_1 .
- Thus the output voltage is $E_0 = ES_1 - ES_2$ and 180° out of phase with primary voltage.
- The amount of voltage change in either secondary winding is proportional to the amount of movement of the core. Hence, we have an indication of amount of linear motion.
- By noticing whether output voltage is increased or decreased, we can determine the direction of motion. The characteristics of LVDT is shown in figure below.



Advantages

- Linearity is good up to 5 mm of displacement.
- Output is rather high. Therefore, immediate amplification is not necessary.
- Output voltage is stepless and hence the resolution is very good.
- Sensitivity is high (about 40 V/mm).
- It does not load the measurand mechanically.
- It consumes low power and low hysteresis loss also.

Disadvantages

- LVDT has large threshold.
- It is affected by stray electromagnetic fields. Hence proper shielding of the device is necessary.

- The ac inputs generate noise.
- Its sensitivity is lower at higher temperature.
- Being a first-order instrument, its dynamic response is not instantaneous.

Applications

- This transducer can also work as a secondary transducer.
- **LVDT is used to measure the weight, force and also pressure.**
- Some of these transducers are used to calculate the pressure and load.
- LVDT's are mostly used in industries as well as servomechanisms.
- Other applications like power turbines, hydraulics, automation, aircraft, and satellites.

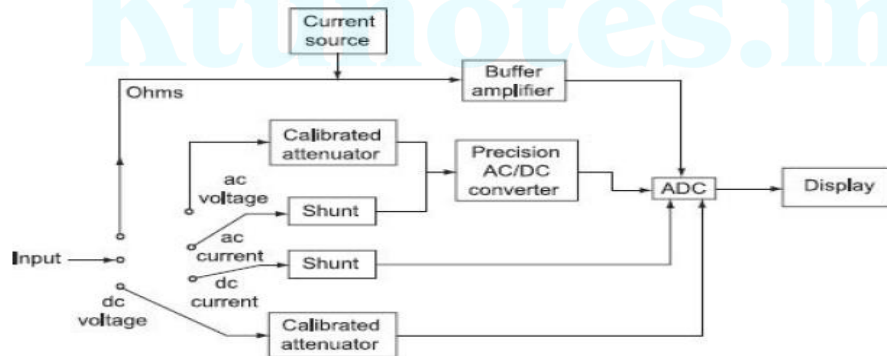
2. Write short notes on

I. Digital Multi Meter and II. Clamp on meter [8 marks] (December 2021)

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Describe Digital Multimeter. [4 marks] (December 2021)

- A digital multimeter is an electronic instrument which can measure very precisely the dc and ac voltage, current (dc and ac), and resistance.
- All quantities other than dc voltage is first converted into an equivalent dc voltage by some device and then measured with the help of digital voltmeter.
- The block diagram of a digital multimeter is shown in [Figure](#).



Block diagram of a digital multimeter

- For measurement of ac voltage, the input voltage, is fed through a calibrated, compensated attenuator, to a precision full-wave rectifier circuit followed by a ripple reduction filter.
- The resulting dc is fed to an Analog Digital Converter (ADC) and the subsequent display system.
- Many manufacturers provide the same attenuator for both ac and dc measurements.
- For current measurement, the drop across an internal calibrated shunt is measured directly by the ADC in the 'dc current mode', and after ac to dc conversion in the 'ac current mode'.
- This drop is often in the range of 200 mV (corresponding to full scale).
- Due to the lack of precision in the ac–dc conversions, the accuracy in the ac range is generally of the order of 0.2 to 0.5%.

- In addition, the measurement range is often limited to about 50 Hz at the lower frequency end due to the ripple in the rectified signal becoming a non-negligible percentage of the display and hence results in fluctuation of the displayed number.
- At the higher frequency end, deterioration of the performance of the ADC converter limits the accuracy.
- In ac measurement the reading is often average or rms values of the unknown current.

II

Parts Of Clamp Meter



- Clamp meter is often known as Tong Tester also.
- It is an easy to use test equipment and is useful in the measurement of a live-conductor without damaging / powering down the circuit.
- One of the biggest advantage of this equipment is that one may take the measurement of a high value current even without powering off/shutting down the circuitry under test.
- The only disadvantage of this test equipment (Clamp meter or Tong tester) is that the accuracy of the tong-tester is considerably low.

Parts of Clamp on Meters

- Generally, a Clamp Meter or Tong Tester has the following parts: –

1. Transformer Clamps (Jaws)

- It senses the magnetic field of the current flowing through the conductor.

2. Clamp Opening Trigger

- It is used to open/ close the clamps.

3. Power Switch

- As the name suggests, it is used to power on/ off the meter.

4. Back Light Button

- It is used to light up the LCD display for easy reading of displayed value in dark places or at night.

5. Hold Button

- It is used to hold the last value on the display.

6. Negative/ Ground Input Terminal

- It is used to connect the negative/ ground jack of the meter cable.

7. Positive Input Terminal

- It is used to connect the positive jack of the meter cable.

8. LCD Display

- It displays the measured value.

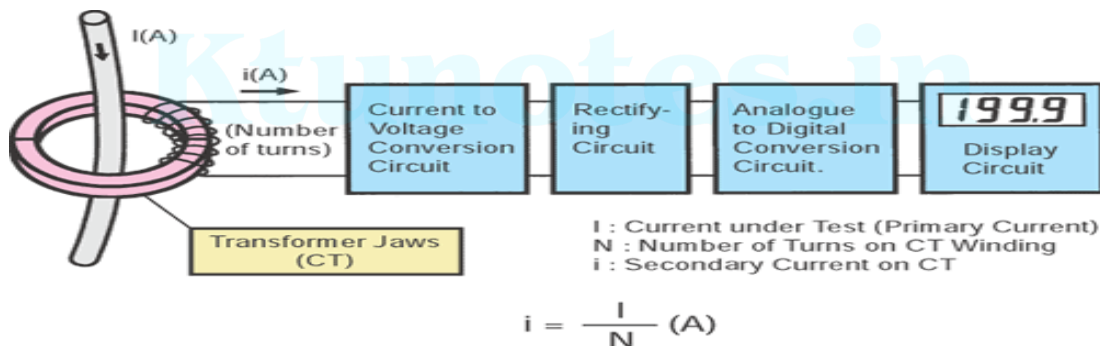
9. Functional Rotary Switch

- It is used to select the type and range of current being measured

In general AC clamp meters operate on the principle of current transformer(CT) used to pick up magnetic flux generated as a result of current flowing through a conductor.

Assuming a current flowing through a conductor to be the primary current, you can obtain a current proportional to the primary current by electromagnetic induction from the secondary side(winding) of the transformer which is connected to a measuring circuit of the instrument.

This permits you to take an AC current reading on the digital display(in the case of digital clamp meters) as illustrated by the block diagram.



- Assume that the conducting material from which the current is passing through is the primary arm of the transformer.
- Due to the current's movement, a magnetic field is generated in it.
- When the arm of tong tester is placed for measurement, it acts as a secondary winding of the transformer.
- The iron core of the arm of tong tester concentrates the magnetic field of the primary winding (i.e. conductor) and hence a current in proportion to the primary-current is generated by electromagnetic-induction.
- This arm of the tong tester is connected to a measurement circuitry which in turn provides the current-reading finally.
- The magnetic field induced in the arms of clamp meter is directly proportional to the number of turns in the secondary winding.

- Since the number of turns in the secondary winding is much more compared to the number of turns in the primary winding wrapped around the core, a very small magnitude of current goes to the measuring unit's input.

3. How strain is measured using a strain gauge? [4marks] (December 2021)

Theory of Strain Gauges. The change in the value of resistance by straining the gauge may be partly explained by the normal dimensional behaviour of elastic material. If a strip of elastic material is subjected to tension, or in other words positively strained, its longitudinal dimension will increase while there will be a reduction in the lateral dimension. So when a gauge is subjected to a positive strain, its length increases while its area of cross-section decreases. Since the resistance of a conductor is proportional to its length and inversely proportional to its area of cross-section, the resistance of the gauge increases with positive strain. The change in the value of resistance of strained conductor is more than what can be accounted for an increase in resistance due to dimensional changes.

The extra change in the value of resistance is attributed to a change in the value of resistivity of a conductor when strained. This property, as described earlier, is known as **piezoresistive effect**.

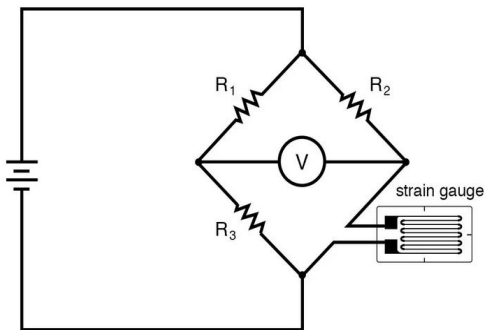
Bridge Measurement Circuit

Typically, the rheostat arm of the bridge (R_2 in the diagram) is set at a value equal to the strain gauge resistance with no force applied.

The two ratio arms of the bridge (R_1 and R_3) are set equal to each other.

Thus, with no force applied to the strain gauge, the bridge will be symmetrically balanced and the voltmeter will indicate zero volts, representing zero force on the strain gauge.

As the strain gauge is either compressed or tensed, its resistance will decrease or increase, respectively, thus unbalancing the bridge and producing an indication at the voltmeter.



4. Draw a neat block diagram of a Cathode Ray Oscilloscope and describe the function of each block in detail. [10marks] (December 2021)

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With neat block diagram explain a general purpose oscilloscope. [10marks] (December 2020)

- CROs are, in fact, very fast X - Y plotters, displaying an input signal versus another signal or versus time.

- The 'stylus' of this 'plotter' is a luminous spot which moves over the display area in response to an input voltage.
- The luminous spot is produced by a beam of electrons striking a fluorescent screen.
- The extremely low inertia effects associated with a beam of electrons enables such a beam to be used following the changes in instantaneous values of rapidly varying voltages.

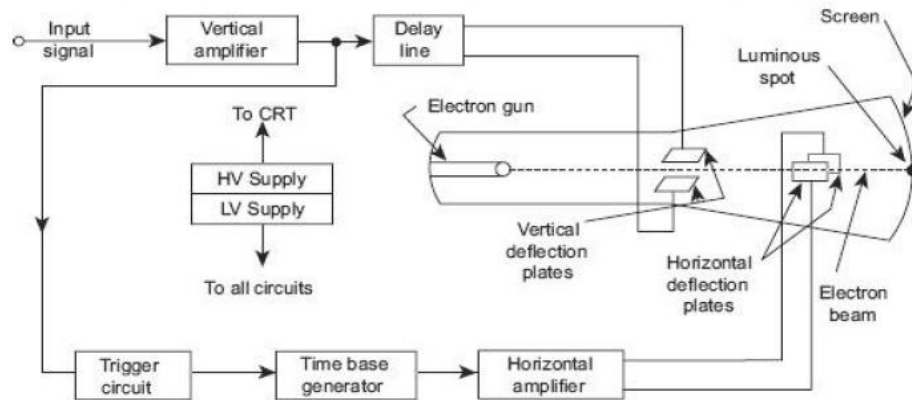


Figure (a) Block diagram of a general-purpose CRO

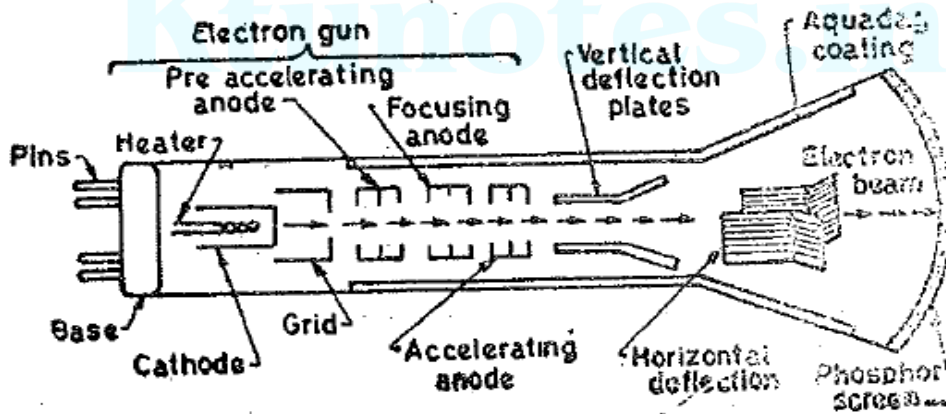


Fig. Internal structure of a CRT.

The main part of the CRO is Cathode Ray Tube (CRT).

It generates the electron beam, accelerates the beam to a high velocity, deflects the beam to create the image and contains a phosphor screen where the electron beam eventually becomes visible.

The phosphor screen is coated with 'aquadag' to collect the secondary emitted electrons.

For accomplishing these tasks, various electrical signals and voltages are required, which are provided by the power supply circuit of the oscilloscope.

Low voltage supply is required for the heater of the electron gun for generation of electron beam and high voltage, of the order of few thousand volts, is required for cathode ray tube to accelerate the beam.

Normal voltage supply, say a few hundred volts, is required for other control circuits of the oscilloscope.

Horizontal and vertical deflecting plates are fitted between the electron gun and screen to deflect the beam according to the input signal.

The main parts of a CRT are

- (i) Electron gun assembly, (ii) Deflection plate assembly, (iii) Fluorescent screen,
- (iv) Glass envelope, (v) Base, through which connections are made to various parts.

The "Electron gun assembly" produces a sharply focused beam of electrons which are accelerated to high velocity. This focused beam of electrons strikes the fluorescent screen with sufficient energy to cause a luminous spot on the screen.

After leaving the electron gun, the

electron beam passes through two pairs of "Electrostatic deflection plates". Voltages applied to these plates deflect the beam. Voltages applied to one pair of plates move the beam vertically up and down and the voltages applied to the other pair of plates move the beam horizontally from one side to another. These two movements i.e. horizontal and vertical are independent of each other and thus the beam may be positioned anywhere on the screen.

The electrons, emitted from the cathode and passing through the hole in the control grid are accelerated by the high positive potential which is applied to the "pre-accelerating" and "accelerating anodes".

The electron beam is focused by the "focusing anode". The accelerating and focusing anodes are cylindrical in form, with small openings located in the centre of each electrode, coaxial with the tube axis. After leaving the focusing anodes, the electron beam passes through the vertical and horizontal deflection plates and then goes on to the fluorescent screen.
